# B. 5-Mark Questions – Exam-Ready Answers

These answers are written in simple, easy-to-learn language for 5 marks each. No figures or diagrams are included, as requested. Each answer contains the key definitions, explanations, examples, and formulas needed for exams.

## 1. Explain the working of SNMP.

SNMP (Simple Network Management Protocol) is an application-layer protocol used to monitor and manage network devices such as routers, switches, and servers.

Components:

1. SNMP Manager: Monitors and controls network devices by sending requests.
2. SNMP Agent: Software running on a managed device that collects and provides device information.
3. MIB (Management Information Base): A database containing information about managed device parameters.

Working:

- The manager sends a GET request to the agent to obtain information.
- The agent retrieves the requested information and sends a RESPONSE.
- The manager uses SET requests to modify device parameters.
- The agent sends a TRAP message to notify the manager about important events, such as link failure.
- SNMP commonly uses UDP port 161 for requests and responses and port 162 for traps.

Example: A network administrator uses SNMP to check router traffic and receive an alert when a network link fails.

## 2. Explain the roles of SNMP, SMI, and MIB in network management.

SNMP, SMI, and MIB work together to monitor and manage network devices.

1. SNMP (Simple Network Management Protocol): Defines how the manager and agent exchange management information through messages such as GET, SET, RESPONSE, and TRAP.
2. SMI (Structure of Management Information): Defines the rules for naming, defining, and representing management information, including data types and object identifiers.
3. MIB (Management Information Base): Stores the definitions of managed objects, such as interface status, packet count, and device uptime.

How they work together:

- SMI defines the structure and format of management objects.
- MIB identifies the managed objects and their properties.
- SNMP allows the manager to access or modify those objects through the agent.

Example: To check a router's interface status, the manager uses SNMP to request the corresponding MIB object, which is defined according to SMI rules.

# 3. Explain how new data types and subtypes are created in ASN.1. Illustrate SEQUENCE and SEQUENCE OF.

ASN.1 (Abstract Syntax Notation One) is a standard notation used to define data structures for communication between systems.

Creating new data types and subtypes:

- A new type can be defined using an existing type.
- A subtype can be created by restricting the values or range of an existing type.
- The keyword `SIZE` can restrict the size of a string or collection.

Examples:

- New type: `Age ::= INTEGER`
- Subtype: `ChildAge ::= INTEGER (0..17)`
- Size restriction: `Name ::= IA5String (SIZE (1..20))`

SEQUENCE: Represents a collection of different fields in a fixed order. Each field may have a different data type.

Example: `Student ::= SEQUENCE { id INTEGER, name IA5String }`

SEQUENCE OF: Represents a list of zero or more elements of the same type.

Example: `Marks ::= SEQUENCE OF INTEGER`

Here, `Student` contains different fields, whereas `Marks` contains multiple integer values.

# 4. Explain analog and digital signals. List the three characteristics of a sine wave.

Analog signal: An analog signal is continuous in time and amplitude. It can take infinitely many amplitude values within a range.

Example: Human voice and traditional radio signals.

Digital signal: A digital signal has discrete amplitude levels and is commonly represented using binary values 0 and 1.

Example: Data transmitted between computers.

Three characteristics of a sine wave:

1. Amplitude: The maximum height of the wave from its zero or reference level. It determines the signal strength.
2. Frequency: The number of cycles completed per second, measured in hertz (Hz).
3. Phase: The relative position of the wave with respect to a reference point, measured in degrees or radians.

The relationship between frequency and period is:

f=1Tf=\frac{1}{T}

where ff is frequency and TT is the time period.

# 5. Explain the three causes of transmission impairment.

Transmission impairment occurs when a signal changes or loses quality while travelling through a communication medium.

1. Attenuation

- Attenuation is the reduction in signal strength as the signal travels through a medium.
- It is measured in decibels (dB).
- Example: A signal becomes weaker when transmitted over a long cable.
- Amplifiers or repeaters can help compensate for signal loss.

2. Distortion

- Distortion occurs when the shape of a signal changes during transmission.
- Different frequency components may experience different delays or attenuation.
- Example: A digital pulse becomes stretched or distorted while travelling through a channel.

3. Noise

- Noise is unwanted energy added to the original signal.
- Common types include thermal noise, induced noise, crosstalk, and impulse noise.
- Example: Electrical interference from a motor affects a nearby communication cable.

These impairments can cause errors and reduce the quality of received data.

# 6. State and explain the Nyquist bit rate formula and Shannon capacity formula. Under what conditions is each applicable?

1. Nyquist bit rate formula

The Nyquist formula gives the maximum theoretical bit rate for a noiseless channel of limited bandwidth.

Rmax=2Blog2LR_{\max}=2B\log_2 L

Where:

- Rmax R_{\max} = maximum bit rate in bits per second
- BB = channel bandwidth in hertz
- LL = number of distinct signal levels

Condition: Applicable to an ideal noiseless channel.

Example: If bandwidth is 3000 Hz and the signal has 4 levels:

Rmax=2(3000)log24R_{\max}=2(3000)\log_2 4
Rmax=12000 bpsR_{\max}=12000\text{ bps}

2. Shannon capacity formula

The Shannon formula gives the theoretical maximum channel capacity in the presence of noise.

C=Blog2(1+SN)C=B\log_2\left(1+\frac{S}{N}\right)

Where:

- CC = channel capacity in bits per second
- BB = bandwidth in hertz
- S/NS/N = signal-to-noise power ratio (linear, not in dB)

Condition: Applicable to a noisy channel with a specified bandwidth and signal-to-noise ratio.

Example: If B=3000B=3000 Hz and S/N=15S/N=15:

C=3000log2(16)=12000 bpsC=3000\log_2(16)=12000\text{ bps}

Difference: Nyquist considers bandwidth and signal levels in a noiseless channel, whereas Shannon considers bandwidth and noise in a noisy channel.

# 7. Explain the steps involved in Pulse Code Modulation (PCM).

Pulse Code Modulation (PCM) is a technique used to convert an analog signal into digital form.

Steps involved:

1. Sampling: The analog signal is measured at regular intervals. According to the sampling theorem, the sampling frequency must be at least twice the highest signal frequency.
2. Quantization: Each sample's amplitude is approximated to the nearest permitted discrete level.
3. Encoding: Each quantized level is represented by a binary code.
4. Transmission: The resulting binary sequence is transmitted through the communication channel.
5. Reconstruction: At the receiver, the digital data is decoded and passed through a reconstruction filter to obtain an approximation of the original analog signal.

Example: PCM is used in digital telephony and audio recording.

For nn bits per sample, the number of quantization levels is:

L=2nL=2^n

The PCM bit rate is:

Rb=nfsR_b=n f_s

where fsf_s is the sampling frequency.

# 8. Explain NRZ-L, NRZ-I, and Manchester line coding schemes for the bit sequence 010011.

Line coding converts digital data into digital signals suitable for transmission.

Consider the bit sequence 010011.

1. NRZ-L (Non-Return-to-Zero-Level)

- The signal level directly represents the data bit.
- One level represents 0, and the other represents 1.
- The signal does not return to zero between bits.
- The exact voltage assigned to each bit depends on the convention used.

Using 0 = low and 1 = high, the levels are:

```
Low, High, Low, Low, High, High
```

2. NRZ-I (Non-Return-to-Zero-Invert)

- A transition at the beginning of a bit interval represents one binary value; no transition represents the other.
- In the common convention, a transition represents 1, and no transition represents 0.
- Assuming the initial level is low, the levels for 010011 are:

```
Low, High, High, High, Low, High
```

NRZ-I is more resistant to polarity inversion than NRZ-L, but long runs without transitions can cause synchronization problems.

3. Manchester coding

- Each bit interval contains a transition in the middle.
- In one common convention, 0 is represented by a low-to-high transition and 1 by a high-to-low transition.
- Every bit contains a clock transition, which helps synchronization.

For 010011, the half-bit transitions are:

- 0: Low → High
- 1: High → Low
- 0: Low → High
- 0: Low → High
- 1: High → Low
- 1: High → Low

The Manchester convention may be reversed in some textbooks.

# 9. Explain the need for block coding. Describe 4B/5B coding and how it solves the synchronization problem of NRZ-I.

Block coding converts groups of data bits into larger groups of code bits before transmission.

Need for block coding:

- It helps maintain synchronization between transmitter and receiver.
- It reduces long sequences without signal transitions.
- It can provide invalid code patterns that help detect certain errors.
- It improves the suitability of data for line coding.

4B/5B coding technique:

- The input data is divided into groups of 4 bits.
- Each 4-bit group is replaced with a 5-bit code.
- The extra bit provides additional transition opportunities.
- A suitable code table is used so that valid codewords avoid long runs of zeros.

Examples:

- `0000` → `11110`
- `0001` → `01001`
- `0010` → `10100`

How it solves the NRZ-I synchronization problem:

NRZ-I depends on transitions to help the receiver recover timing. A long sequence of zeros may contain no transitions, making synchronization difficult. The 4B/5B technique limits long runs of zeros, helping ensure enough transitions in the encoded stream.

Note: 4B/5B adds overhead because every 4 data bits become 5 transmitted bits.

# 10. Explain ASK, FSK, and PSK with examples and bandwidth formulas.

These are digital-to-analog modulation techniques used to transmit digital data over analog communication channels.

1. Amplitude Shift Keying (ASK)

- The amplitude of the carrier changes according to the input data.
- The frequency and phase remain constant.
- Example: Binary 1 uses a carrier of larger amplitude, while binary 0 uses a smaller amplitude or no carrier.
- It is simple but sensitive to amplitude noise.

Approximate bandwidth for binary ASK:

B≈(1+d)RsB\approx (1+d)R_s

2. Frequency Shift Keying (FSK)

- The carrier frequency changes according to the input bits.
- The amplitude generally remains constant.
- Example: One frequency represents 0 and another represents 1.
- It has better noise immunity than ASK in many applications.

For binary FSK, a common approximate bandwidth expression is:

B≈|f1−f0|+(1+d)RsB\approx |f_1-f_0|+(1+d)R_s

3. Phase Shift Keying (PSK)

- The phase of the carrier changes according to the input data.
- The carrier amplitude and frequency remain constant.
- Example: In BPSK, two phases separated by 180° represent binary 0 and 1.

Approximate bandwidth for BPSK:

B≈(1+d)RsB\approx (1+d)R_s

Here, RsR_s is the symbol rate and dd is the roll-off factor in a raised-cosine-shaped system. Exact bandwidth depends on the pulse shaping and modulation scheme.

# 11. Explain QAM and why it is more spectrally efficient than ASK or PSK.

Quadrature Amplitude Modulation (QAM) is a modulation technique that changes both the amplitude and phase of a carrier to represent digital data.

Working:

- QAM combines two carrier signals of the same frequency that are 90° apart in phase.
- These are called the in-phase (I) and quadrature (Q) components.
- By varying both components, QAM produces different signal states.
- Each state represents a group of bits.

Examples: 16-QAM represents 4 bits per symbol, while 64-QAM represents 6 bits per symbol.

The number of bits per symbol is:

k=log⁡2Mk=\log_2 M

where MM is the number of distinct signal states.

Why QAM is more spectrally efficient:

1. It transmits multiple bits in each symbol.
2. It uses both amplitude and phase to represent more states.
3. It can achieve a higher data rate within the same bandwidth than basic binary ASK or BPSK.
4. It is widely used in Wi-Fi, cable modems, and broadband communication.

Limitation: Higher-order QAM requires a better signal-to-noise ratio because its signal states are closer together.

# 12. Explain AM, FM, and PM. Tabulate what changes and what remains constant.

Analog modulation changes a carrier signal according to the message signal.

1. Amplitude Modulation (AM): The amplitude of the carrier changes according to the message signal. Its frequency and phase remain constant for an ideal carrier.

2. Frequency Modulation (FM): The instantaneous frequency of the carrier changes according to the message signal. Its amplitude remains constant.

3. Phase Modulation (PM): The instantaneous phase of the carrier changes according to the message signal. Its amplitude remains constant.

| Feature | AM | FM | PM |
|---|---|---|---|

| Parameter that changes | Amplitude | Frequency | Phase |
|---|---|---|---|
| Amplitude | Changes | Constant | Constant |
| Frequency | Carrier frequency remains fixed in the basic model | Changes instantaneously | Can vary as phase changes |
| Phase | Carrier phase remains fixed in the basic model | Varies with the integral of the message | Changes directly with the message |
| Noise immunity | Lower | Generally higher | Generally higher |
| Typical application | AM broadcasting | FM radio | Communication and modulation systems |

# 13. Explain FDM and TDM. Differentiate between synchronous and statistical TDM.

1. Frequency Division Multiplexing (FDM)

FDM combines multiple signals and transmits them simultaneously over one communication channel by assigning each signal a separate frequency band.

- Each source uses a different carrier frequency.
- Guard bands may be used to reduce interference between adjacent channels.
- It is commonly used in radio broadcasting and cable television.

2. Time Division Multiplexing (TDM)

TDM allows multiple sources to share one communication channel by assigning them different time slots.

- Each source transmits during its allocated time interval.
- The receiver separates the signals using timing information.
- It is commonly used in digital communication systems.

Synchronous TDM vs Statistical TDM

| Synchronous TDM | Statistical TDM |
|---|---|
| Each input gets a fixed time slot. | Time slots are assigned as needed. |
| Slots may be wasted if a source has no data. | Bandwidth is used more efficiently. |
| Simple to implement. | Requires buffering and addressing or identification information. |
| Timing and slot positions are predetermined. | Slot allocation depends on active sources. |
| Used in traditional digital telephone systems. | Used in packet-oriented data communication. |

# 14. Explain FHSS and DSSS spread spectrum techniques with examples.

Spread spectrum is a communication technique in which a signal is spread over a wider frequency band than the minimum required for its data rate.

1. Frequency Hopping Spread Spectrum (FHSS)

- The transmitter rapidly changes its carrier frequency according to a predetermined pseudorandom sequence.
- The receiver follows the same hopping sequence.
- Interference affecting one frequency may have limited impact on the whole communication.
- Example: Classic Bluetooth uses frequency hopping.

Advantages: Resistance to narrowband interference and some forms of jamming.

2. Direct Sequence Spread Spectrum (DSSS)

- The transmitter combines each data bit with a faster pseudorandom chip sequence.
- This spreads the signal across a wider bandwidth.
- The receiver uses the same chip sequence to recover the original data.
- Example: IEEE 802.11b Wi-Fi supports DSSS-based techniques.

Advantages: Improved resistance to narrowband interference and the ability to recover data through correlation with the spreading code.

Difference: FHSS spreads the signal by changing carrier frequencies, whereas DSSS spreads it by multiplying or combining the data with a high-rate chip sequence.

# 15. Compare twisted-pair, coaxial, and fiber-optic cables.

Transmission media carry data between devices in a network.

| Feature | Twisted-pair cable | Coaxial cable | Fiber-optic cable |
|---|---|---|---|
| Construction | Two insulated copper wires twisted together | Central conductor surrounded by insulation and shielding | Glass or plastic fibers that carry light |
| Bandwidth | Moderate to high, depending on category | Moderate to high | Very high |
| Cost | Generally low | Moderate | Higher installation cost in many cases |
| Noise immunity | Moderate; better with shielding | Good due to shielding | Excellent; immune to electromagnetic interference |

| | | | |
|---|---|---|---|
| Signal attenuation | Relatively high over long distances | Lower than basic twisted pair in many applications | Low over long distances |
| Distance | Usually limited for Ethernet copper links | Moderate | Suitable for very long distances |
| Installation | Easy | More difficult than twisted pair | Requires specialized tools and handling |
| Applications | LANs and telephone lines | Cable TV and broadband | Backbone networks and long-distance communication |

Conclusion: Twisted pair is economical for ordinary LANs, coaxial cable provides good shielding, and fiber-optic cable is preferred for very high bandwidth and long-distance transmission.

# 16. Explain radio waves, microwaves, and infrared as unguided transmission media.

Unguided transmission media transmit electromagnetic signals through air or space without a physical cable.

1. Radio waves

- Radio waves are generally omnidirectional, meaning they can spread in many directions.
- They can travel over long distances, and some frequency bands can penetrate buildings.
- They may be affected by interference from other transmitters.
- Applications: AM/FM broadcasting, radio communication, and some wireless networks.

2. Microwaves

- Microwaves are commonly used for directional, point-to-point communication.
- They often require line-of-sight alignment between transmitting and receiving antennas.
- They provide high data rates and are suitable for terrestrial links and satellite communication.
- Applications: Satellite communication, cellular networks, and microwave relay links.

3. Infrared

- Infrared communication is usually short-range and directional or confined to a room.
- It generally does not penetrate walls, which can reduce interference between rooms.
- It is commonly used for short-distance communication.
- Applications: TV remote controls, infrared sensors, and some short-range device links.

Comparison: Radio waves are useful for broad-area coverage, microwaves are suitable for directional high-capacity links, and infrared is mainly used for short-range communication.

# Quick revision tips for 5 marks

- Write the definition first.
- Include 3–5 key points for each topic.
- Memorize the formulas in Questions 6, 7, and 10.
- For comparison questions, remember the important differences in table form.
- Add one practical example wherever possible.
- Since you don't need figures, focus on keywords, working principles, formulas, advantages, and applications.